arXiv:2508.06988v1 [cs.CV] 9 Aug 2025

TADoc: Robust Time-Aware Document Image Dewarping

Fangmin Zhao[a,c], **Weichao Zeng**[a,c], **Zhenhang Li**[a,c], **Dongbao Yang**[a] **and Yu Zhou**[b,*]

[a]Institute of Information Engineering, Chinese Academy of Sciences
[b]VCIP & TMCC & DISSec, College of Computer Science & College of Cryptology and Cyber Science, Nankai University
[c]School of Cyber Security, University of Chinese Academy of Sciences

Abstract. Flattening curved, wrinkled, and rotated document images captured by portable photographing devices, termed document image dewarping, has become an increasingly important task with the rise of digital economy and online working. Although many methods have been proposed recently, they often struggle to achieve satisfactory results when confronted with intricate document structures and higher degrees of deformation in real-world scenarios. Our main insight is that, unlike other document restoration tasks (e.g., deblurring), dewarping in real physical scenes is a progressive motion rather than a one-step transformation. Based on this, we have undertaken two key initiatives. Firstly, we reformulate this task, modeling it for the first time as a dynamic process that encompasses a series of intermediate states. Secondly, we design a lightweight framework called TADoc (**T**ime-**A**ware **Doc**ument Dewarping Network) to address the geometric distortion of document images. In addition, due to the inadequacy of OCR metrics for document images containing sparse text, the comprehensiveness of evaluation is insufficient. To address this shortcoming, we propose a new metric – DLS (**D**ocument **L**ayout **S**imilarity) – to evaluate the effectiveness of document dewarping in downstream tasks. Extensive experiments and in-depth evaluations have been conducted and the results indicate that our model possesses strong robustness, achieving superiority on several benchmarks with different document types and degrees of distortion.

1 Introduction

Document images captured by mobile photography devices suffer from various distortions and background interference due to uncontrolled paper deformations and camera position. These drawbacks not only affect the users' reading experience but also impede downstream tasks (e.g., layout analysis [17], optical character recognition [43, 35, 26, 32, 21], document question answering[27]), wherein expert models are mostly trained on scanned documents. Therefore, document image dewarping, aimed at eliminating geometric distortions, has received increasing attention. Traditional dewarping methods focus mainly on measuring the 3D shape of degraded documents. They often rely on carefully designed and calibrated hardware, such as stereo cameras and structured light projectors [24], which suffer from a vast lack of practicality. Other methods leverage manually designed low-level features such as shading [41] and contours, yet limited performance has been achieved.

* Corresponding Author. Email: yzhou@nankai.edu.cn.

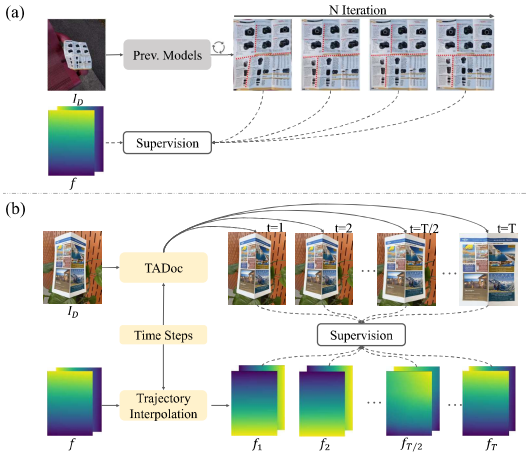

Figure 1. Comparison of TADoc with previous progressive methods. For distorted image I_D: (a) Previous methods are trained with single mapping supervision f with cascaded network. (b) TADoc provides multi-step prediction and leverages dynamic distortion mapping f_t for fine-grained supervision.

Recently, methods based on deep learning have been proposed for document dewarping. These works [22, 5, 33] typically model the task as a regression problem with a 2D vector field that represents the coordinate flow between distorted and flat images. Some methods have been proposed to flatten documents utilizing prior information such as 3D [34], text lines [12] and layout [19]. However, they require either additional complex annotations or time-consuming post-processing operations. Meanwhile, these models may suffer from degeneration when faced with a document with sparse textlines or unclear layout. As shown in Fig. 2, previous methods fail to achieve satisfactory overall results on multiple datasets.

In this work, inspired by the scenario of flattening a distorted document in the real world, we reformulate the document dewarping task as a progressive rectification process. As shown in Fig. 1 (b), we divide document dewarping into T time steps, and as the time step gradually increases from 0 to T, the document gradually recovers from the original distorted image to a completely flat image. This modeling has distinct advantages over previous progressive methods [40], as shown in Fig. 1 (a). Firstly, previous methods use the results from the previous step as input, achieving a coarse-to-fine dewarping effect, which can lead to imbalanced learning difficulty across different stages and requires substantial time due to its sequential execution. In contrast, our method consistently takes the original distorted image as input. Given a time step, we compel the model to

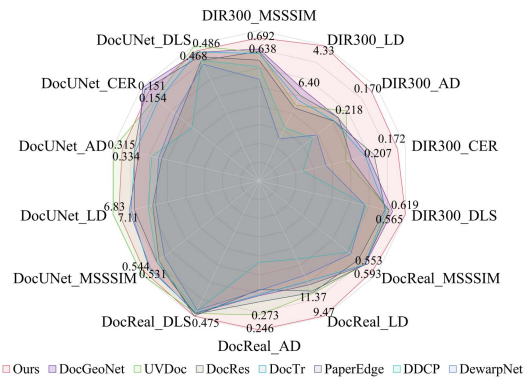

Figure 2. The performance comparison on several benchmarks and metrics.

learn the corresponding degree of distortion, thereby enhancing its understanding of the distortion process and leading to superior performance. Additionally, our method supports parallel inference, with significantly higher processing speeds.

Based on multi-step temporal modeling, a lightweight model called TADoc (**T**ime-**A**ware **Doc**ument Dewarping Network) is proposed. Specifically, we introduce time embedding to extract time-aware distorted document features explicitly. Then, we decode the features and apply the predicted coordinate map to the original image to obtain different degrees of restored images. Fig. 2 displays the superiority of TADoc, with strong robustness to a wide range of document types and distortion levels.

In addition, previously OCR metrics for downstream evaluation fail to evaluate documents with sparse texts or no texts, leading to insufficient evaluation. To address the limitation, we propose a supplementary metric for evaluating the performance of document dewarping, termed Document Layout Similarity (DLS). Specifically, a pre-trained document layout analysis (DLA) model is leveraged to process the dewarped document image and the corresponding scanned one, while DLS is defined as the Intersection-over-Union (IoU) of the analysis results. Compared with OCR, DLA is not only closely related to the users' reading experience, but can also be applied to all document images for comprehensive evaluation, thus better reflecting the quality of dewarping from a downstream perspective.

Overall, we have made three-fold contributions in this work for document image dewarping as follows:

- Inspired by the temporal progress of flattening in real-world scenarios, we reformulate document dewarping as a progressive process. To our knowledge, we are the first to introduce the concept of time steps into document image dewarping.
- Based on the reformulation, we proposed a lightweight TADoc, which benefits from the temporal modeling to capture subtle distortion and achieves state-of-the-art performance on several commonly used benchmarks.
- To alleviate the shortcoming of OCR assessment, we propose a new metric – Document Layout Similarity (DLS), provides a comprehensive evaluation of dewarping results in downstream tasks.

2 Related Works

2.1 Traditional Document Dewarping

Traditional dewarping methods mainly focus on 3D reconstruction and 2D prior. Multi view images or additional hardwares are often used to assist in 3D reconstruction. Zhang et al. [41] use a range/depth tensor to measure the 3D shape of the deformed papers. Meng et al. [24] obtain the document curl using a platform with two structured laser beams. Tsoi and Brown [30] synthesize the flat image by combining the boundary of images from different perspectives. You et al. [36] model the crease information on multiple images and reconstruct the 3D shape. For 2D image processing, low-level features such as text lines are often emphasized. Mischke and Luther [25] model the detected text lines as polynomial curves, and Lavialle et al. [15] choose to model them as cubic B-splines.

2.2 Deep Learning-Based Document Dewarping

With the development of deep learning, many document dewarping methods [22, 5, 14, 28, 20, 39] based on deep learning have been proposed. DocUNet [22] and DocTr [5] use CNN and Transformer architectures, respectively, to predict the global map. DewarpNet [3] and CREASE [23] first predicts the 3D coordinates, and then generates a dense backward map. DDCP [33], DocReal [38] and UVDoc [31] proposes to predict sparse control point coordinates. Methods such as DocGeoNet [6] and La-DocFlatten [19] utilize prior information like edges, text lines, and layout to improve the dewarping effects. Marior [40] proposes to dewarp images from coarse to fine. Tang et al. [29] proposed a dual-stream architecture that addresses both geometric distortion and uneven illumination simultaneously, while Kumari et al. [14] introduced the concept of transfer learning, applying geometric correction from natural scenes to the document domain. Zhang et al. [42] train image registration models to obtain real images with annotations, thereby improving dewarping effects from a data perspective. DocMamba [9] achieves robust dewarping effect based on an selective state space sequence model.

2.3 Document Layout Analysis

DLA aims to recognize document layout and position. Lebourgeois et al. [16] connect the characters into blocks, and then connect them into paragraphs. Journet et al. [13] distinguish text and graphic regions by extracting directional features within smaller square areas. Diem et al. [4] use clustering to classify document regions. Garz et al. [7] use SIFT features to detect layouts. The deep learning approaches model DLA as a detection [18] or segmentation task [1] using CNN or transformer frameworks, benefiting from the development of Faster-RCNN [8], Mask-RCNN [11], and others.

3 Proposed Method

Document dewarping in a progressive manner naturally coincides with the temporal progress of flattening in real-world scenarios. In this section, we will reformulate the problem, then detail the framework, TADoc, followed by the paradigm of training and inference.

3.1 Problem Reformulation

Given a distorted document image I_D, we obtain the restored image I_R^t at each time step t through the network by constructing the corresponding warping flow $\mathcal{F}_t$:

$$I_R^t = \mathcal{F}_t(I_D), \tag{1}$$

where $\mathcal{F}_t = \{F^t(p) \mid p \in I_D\}$, p is the pixel coordinate in the distorted document I_D and the mapping $F^t(p)$ at each pixel coordinate p from I_D defines the position of that pixel at each time step t.

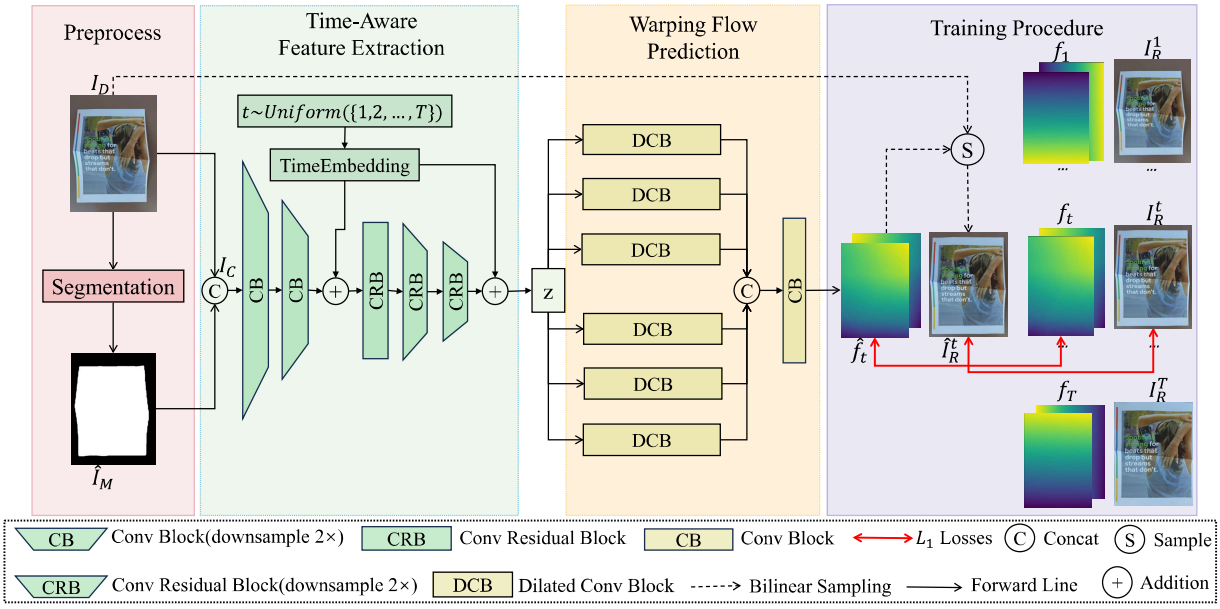

Figure 3. An overview of the proposed TADoc. Three modules are contained, namely *Preprocess*, *Time-Aware Feature Extraction*, and *Warping Flow Prediction*. Moreover, the training paradigm is also briefly illustrated.

To further clarify, at time step t, (x, y) and (u_t, v_t) are the 2D coordinates of restored document I_R^t and distorted document I_D, respectively. The relationship between two coordinates can be represented via a backward map f_t:

$$f_t(x, y) = (u_t, v_t), \tag{2}$$

When the restoration process is divided into T time steps, at time $t = 0$, the document image I_R^0 is identical to I_D, with the corresponding backward map f_0 as:

$$f_0(x, y) = (x, y), \tag{3}$$

wherein f_0 is a regular grid.

At time $t = T$, the restored image I_R^T is a completely flat image, with f_T written as:

$$f_T(x, y) = (u_T, v_T), \tag{4}$$

wherein f_T is equivalent to the backward map annotations.

For an intermediate time step t, the degree of image restoration varies from f_0 to f_T. For simplicity, we consider a linear interpolation to model the trajectory of transport between f_0 and f_T:

$$f_t = \frac{t}{T} f_T + (1 - \frac{t}{T}) f_0. \tag{5}$$

3.2 *Time-Aware Document Dewarping Network*

With the problem reformulated, a time-aware dewarping network is proposed, termed TADoc. It consists of Preprocessing Module, Time-Aware Feature Extraction (TAFE) Module and Warping Flow Prediction (WFP) Module, as depicted in Fig. 3. We will delve into these modules in the following.

Preprocess. To eliminate the extra background texture from the distorted image $I_D \in \mathbb{R}^{H \times W \times 3}$, a segmentation network DeepLabv3+ [2] is adopted to obtain the mask image $I_M \in \mathbb{R}^{H \times W}$. The auxiliary mask guides the model to focus on the content of informative areas. Subsequently, we feed the original image I_D along with the mask image I_M into our proposed model to predict grid map. The preprocessing is fast and doesn't impact the inference speed.

Time-Aware Feature Extraction. The input of TAFE module is $I_C \in \mathbb{R}^{H \times W \times 4}$ and timestep t, wherein I_C is obtained by concatenating I_D with I_M. Compared with pixel-wise multiplication [5], the concatenation allows an adaptive integration of region information while alleviating the negative impact from imprecise masks.

In TAFE, I_C is first fed into two convolutional layers to extract preliminary image information. Timestep t is encoded as a time embedding and fused with preliminary image features. This is followed by multiple layers of residual blocks [10] to get deeper document distortion features. Subsequently, this feature is added to the time embedding to enhance TADoc's ability to control the degree of distortion to obtain the time-aware distortion features $z \in \mathbb{R}^{\frac{H}{16} \times \frac{W}{16} \times c}$.

Warping Flow Prediction. The WFP module aims to use the time-aware features z obtained above to predict the warping flow. Inspired by DDCP [33], we choose to predict the backward map of sparse grid points, which makes our model lightweight without the need for upsampling operations. Specifically, z go through six convolutional blocks with different dilation rates [37] respectively to enrich the network's insight ability into multi-scale and improve robustness. Subsequently, these features are aggregated and sent to a single-layer convolutional layer to obtain the warping flow $f_t \in \mathbb{R}^{\frac{H}{16} \times \frac{W}{16} \times 2}$, which can be seen as the positions of $\frac{H}{16} \times \frac{W}{16}$ reference points in I_D. Finally, f_t is bilinearly interpolated to the size of I_D to obtain the restored document image I_R^t at time t.

Training Loss. During the training stage, TADoc is optimized with the following objective,

$$\mathcal{L} = \alpha ||f_t - \hat{f}_t|| + \beta \mathcal{L}_r. \tag{6}$$

where f_t and $\hat{f}_t$ are the annotated flow and the predicted counterparts. $\mathcal{L}_r$ is L_1 losses on the ground truth flat images and the dewarped images obtained through the predicted grid backward map $\hat{f}_t$. α and β are leveraged to balance the influence of the individual loss terms.

Algorithm 1 Training

Input: Dataset $\mathcal{S}$
Initiate: Model $\mathcal{M}_\theta$

repeat
  $t \sim \text{Uniform}(\{1, 2, \ldots, T\})$
  $I_D, f_T, I_R^t \sim \mathcal{S}$
  $f_t = \frac{t}{T} f_T + (1 - \frac{t}{T}) f_0$
  $\hat{f}_t \leftarrow \mathcal{M}_\theta(I_D, t)$
  $\hat{f}_T = \frac{T}{t} \hat{f}_t + (1 - \frac{T}{t}) f_0$
  $\hat{I}_R^t \leftarrow Sampling(I_D, \hat{f}_T)$
  Take gradient descent step on ∇L
until converged

Algorithm 2 Parallel Inference

Input: Image I_D
Parameter: Trained Model $\mathcal{M}_\theta$

for all $t = 1, 2, ..., T$ in parallel **do**
  $\hat{f}_t \leftarrow \mathcal{M}_\theta(I_D, t)$
  $\hat{f}_T^t = \frac{T}{t} \hat{f}_t + (1 - \frac{T}{t}) f_0$
end for
$\hat{f}_T^{sum} \leftarrow \sum_{t=1}^{T} \hat{f}_T^t$
$f_T^{direct} = \hat{f}_T^T$
$f_T^{avg} = \frac{1}{T} \hat{f}_T^{sum}$
$I_R^{direct} \leftarrow Sampling(I_D, f_T^{direct})$
$I_R^{avg} \leftarrow Sampling(I_D, f_T^{avg})$

3.3 Paradigm of Training and Inference

Now we discuss about the training and inference processes.

Algorithm 1 displays the complete training procedure. For each sample in the training set, we obtain distorted image I_D, ground truth backward map f_T, and ground truth flattened image I_R, and then randomly select a time step t from 1 to T with equal probability to calculate the corresponding supervised signal f_t. Subsequently, we train the model until convergence, enabling it to have a more comprehensive understanding of the dewarping process.

During the inference phase, as shown in Algorithm 2, we try two inference methods: direct method and average method. For direct method, we can simply input the original image I_D and the final time step $t = T$ into the network to predict the backward map f_T^{direct}. For average method, the predicted information from all T time steps is adopted. Specifically, for every time step t, the trained TADoc predicts the backward map $\hat{f}_t$, and the corresponding final map $\hat{f}_T^t$ can be deduced following Eq. 5:

$$\hat{f}_T^t = \frac{T}{t} \hat{f}_t + (1 - \frac{T}{t}) f_0, \tag{7}$$

The result f_T^{avg} of the average method is the average of the final flow obtained for all time steps:

$$f_T^{avg} = \frac{1}{T} \sum_{t=1}^{T} \hat{f}_T^t. \tag{8}$$

Finally, the flattened image I_R^{direct} and I_R^{avg} can be obtained by sampling from I_D through f_T^{direct} and f_T^{avg}.

4 Experiments

4.1 Dataset

TADoc is trained on the Doc3D and UVDoc datasets and evaluated on the DIR300, DocReal and DocUNet benchmarks. A brief introduction of each dataset is provided.

Training Data. The Doc3D dataset contains 100K synthetic distorted document images with rich annotations, including 3D maps, surface normals, albedo maps, depth maps, and backward map. The UVDoc dataset contains 20K pseudo-photorealistic document images with 4,032 geometric deformations. Annotations involve sparse backward maps, sparse 3D grids and masks.

Evaluation Data. The DIR300 benchmark contains 300 real-world document photos captured by different mobile devices with various distortion types. The DocReal benchmark contains 200 Chinese document images, covering a wide range of real-world scenarios involving tables, contracts, books and so on. The DocUNet benchmark contains 130 images from 65 different documents covering receipts, letters, magazines and music sheets.

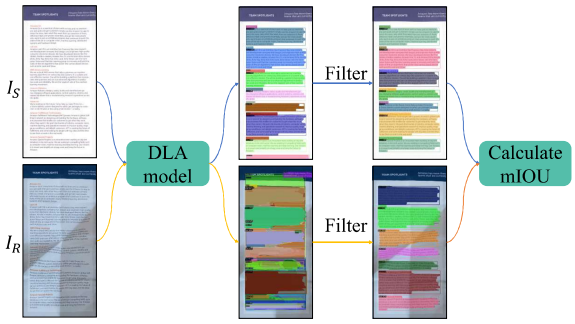

Figure 4. The pipeline of calculating DLS metric.

4.2 Evaluation Metrics

To ensure a comprehensive assessment of dewarping results, image similarity metrics, OCR metrics and the proposed DLS are adopted for quantitative evaluation in the experiments.

Image Similarity Metrics. MS-SSIM builds a Gaussian pyramid of two images and is defined as the weighted sum of SSIM across multiple resolutions. We follow the parameters in DocUNet [22]. Local Distortion (LD) utilizes SIFT flow for dense image registration between two images, measuring the average deformation of each pixel. Aligned Distortion (AD) enhances MS-SSIM through unified translation and scaling operations while alleviating LD errors using image gradients as weighting factors.

OCR Metrics. Edit Distance (ED) is defined as the minimal number of substitutions (s), insertions(i), and deletions (d) to transform a string into the reference one. And Character Error Rate (CER) is defined as $CER = (s + i + d)/N$, where N refers to the total number of characters in the reference text. 90 pairs of images in DIR300, 60 pairs of images in DocUNet and all images in DocReal are generally selected for OCR evaluation, consistent with previous experiments. The selected images contain a great amount of text, contributing to stable result of CER. We perform the evaluation with Tesseract v5.0.1 and PyTesseract v0.3.9.

Document Layout Similarity. Although OCR metrics are important indicators for evaluating the performance of dewarping results in downstream tasks, it cannot process documents with sparse texts or no texts (e.g., music sheets, illustrations). As mentioned above, only a subset of images from the benchmarks are qualified for evaluation, cannot comprehensively reflect the dewarping effects of all document types. Therefore, we turn to another downstream task with looser constraints, DLA. Based on DLA, we propose the DLS. As

Table 1. Quantitative comparison of dewarping performance on the DIR300 benchmark. "↑" and "↓" signifies higher and lower better respectively. The **best** results are represented in bold, the second best results are represented in underline, the third best results are represented in double underline, and "-" indicates that the dewarped results are inaccessible due to lack of open-source availability. The same applies to the tables below.

Method	Venue	MS-SSIM ↑	LD ↓	AD ↓	CER ↓	ED ↓	DLS ↑	Parameter ↓	FPS ↑
DewarpNet	*ICCV'19*	0.492	13.94	0.331	0.356	1059.6	0.447	86.9M	0.43
DocTr	*MM'21*	0.616	7.21	0.254	0.224	699.6	0.537	26.9M	0.64
DDCP	*ICDAR'21*	0.552	10.95	0.357	0.541	2085.0	0.453	13.3M	1.82
PaperEdge	*SIGGRAPH'22*	0.584	8.00	0.255	0.533	508.7	0.324	36.6M	0.50
DocGeoNet	*ECCV'22*	0.638	6.40	0.242	0.565	665.0	0.375	24.8M	0.59
UVDoc	*SIGGRAPH'23*	0.622	7.73	0.218	0.275	675.4	0.554	8.0M	0.87
LA-DocFlatten	*TOG'23*	0.652	5.70	0.195	0.189	511.1	0.589	-	-
DocRes	*CVPR'24*	0.626	6.83	0.241	0.257	763.2	0.549	15.2M	0.44
DocTLNet	*IJDAR'24*	0.658	5.75	-	0.177	482.6	-	-	-
DocMamba	*ICMM'25*	0.632	6.57	0.194	-	-	-	-	-
Ours (direct)		**0.692**	4.39	0.175	0.180	499.0	**0.621**	7.9M	0.78
Ours (average)		**0.692**	**4.33**	**0.170**	**0.172**	**475.0**	0.619	7.9M	0.78

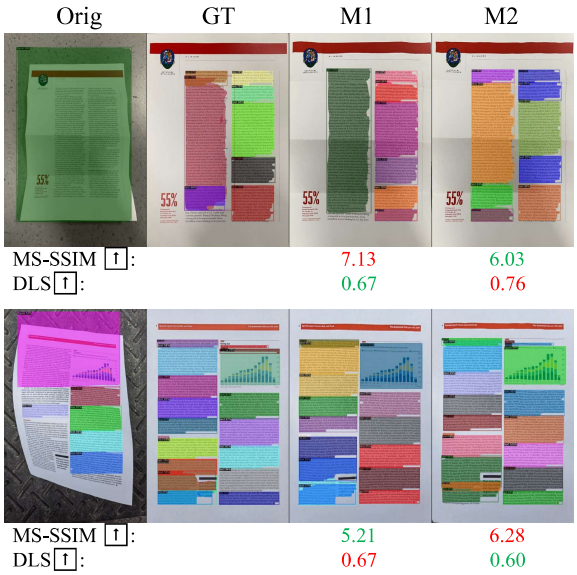

Figure 5. Comparison of DLS and MS-SSIM result.

shown in Fig. 4, we leverage a pretrained model DiT to process scanned document I_S and dewarped document I_R pair, obtaining corresponding layout segmentation regions and categories. Layout areas with a low confidence level are filtered for stability and credibility. Subsequently, for each corresponding layout area pair, the IOU is calculated. DLS represents the average value of IOU across all layout areas, which introduces a novel perspective to evaluate dewarping methods. In addition, while both DLS and MS-SSIM measure the similarity between two images, DLS places greater emphasis on high-level semantic similarity, whereas MS-SSIM focuses more on low-level similarity. Specifically, for identical layout regions in two images, if the DLA model classifies them into different categories, the DLS score will be zero. Overall, DLS imposes stricter requirements on semantic similarity. As shown in Fig. 5, for the dewarping result of two method M1 and M2, they cannot always perform optimally on both MS-SSIM and DLS. Although the result of M1 attains higher MS-SSIM, the result of M2 demonstrates better dewarping effects, particularly in text lines, and also achieves higher DLS score. Therefore, we believe that DLS serves as a supplementary metric and the introduction of DLS is both reasonable and necessary.

4.3 Implementation Details

We implement our model in PyTorch and train it on a NVIDIA GeForce RTX 4090. A batch size of 96 is used with the AdamW optimizer. The initial learning rate is set as 10^{-4} and weight decay is set of 10^{-4}. The hyperparameters α and β of the loss function are both set to 1. During the evaluation, we test FPS of different methods, all on a NVIDIA GeForce RTX 4090.

4.4 Comparison with Prior Methods

We quantitatively and qualitatively compare TADoc with deep learning based document dewarping approaches that have been open-sourced or provide available official results on the DIR300, DocReal and DocUNet benchmarks. Notably, we report the results of our model with both direct method and average method at a total time step of 20.

Quantitative evaluation. As shown in Tab. 1 and Tab. 2, TADoc demonstrate superior overall performance. Specifically, on the DIR300, TADoc not only achieves the best results across all evaluation metrics but also significantly outperforms previous SOTA methods. Notably, the LD and AD metrics show substantial improvements, with reductions of 1.33 and 0.025, respectively. This also demonstrates that our dewarping results not only enhance the users' reading experience but also improve the performance of document images in downstream analysis tasks, including OCR and DLA. On the DocReal, TADoc surpasses previous SOTA on all metrics except AD and ED, where it ranks second, demonstrating robust results. On the DocUNet, we rank second in MS-SSIM, rank third in ED and DLS. In general, benefiting from the temporal reformulation and modeling, TADoc exhibits state-of-the-art or comparable performance on all benchmarks with various document types and deformation degrees.

Notably, no method has achieved a complete advantage on DocUNet. Apart from the OCR metric, which could not be evaluated for all images, we investigate the reasons related to image similarity. As observed by DocGeoNet [6], the reference images in DocUNet may suffer from imperfect alignment due to scanning error, which can lead to significant performance degradation, particularly affecting the LD metric. As shown in Fig. 8 (a), TADoc performs well in restoring document edges and text lines but its quantitative metrics are notably poor. This misalignment issue also affects the comparison of various methods. As shown in Fig. 8 (b), TADoc performs

Table 2. Quantitative comparison of dewarping performance on the DocReal and DocUNet benchmark.

		DocReal						DocUNet					
Method	Venue	MS-SSIM ↑	LD ↓	AD ↓	CER ↓	ED ↓	DLS ↑	MS-SSIM ↑	LD ↓	AD ↓	CER ↓	ED ↓	DLS ↑
DewarpNet	*ICCV'19*	0.480	13.54	0.334	0.781	561.2	0.454	0.439	9.63	0.455	0.210	525.5	0.420
DocTr	*MM'21*	0.551	12.74	0.322	0.801	519.2	0.463	0.510	7.79	0.368	0.175	464.8	0.464
DDCP	*ICDAR'21*	0.468	16.32	0.450	0.961	721.2	0.458	0.473	8.98	0.426	0.263	745.4	0.434
PaperEdge	*SIGGRAPH'22*	0.519	11.45	0.303	0.733	507.3	0.468	0.473	7.99	0.366	0.154	375.6	0.447
Marior	*MM'22*	-	-	-	-	-	-	0.478	7.27	0.403	0.214	593.8	0.421
DocGeoNet	*ECCV'22*	0.553	12.29	0.319	0.733	535.1	0.466	0.505	7.68	0.381	0.151	379.0	0.463
DocReal	*WACV'23*	0.556	9.83	**0.238**	0.758	**466.7**	0.473	0.498	7.83	0.311	**0.144**	**360.3**	0.463
LA-DocFlatten	*TOG'23*	-	-	-	-	-	-	0.526	**6.72**	**0.300**	0.153	391.9	0.476
UVDoc	*SIGGRAPH'23*	0.550	11.37	0.273	0.732	503.5	0.466	**0.544**	6.83	0.315	0.170	461.1	**0.486**
DocRes	*CVPR'24*	0.551	11.61	0.337	0.742	550.8	**0.475**	0.464	9.37	0.470	0.220	598.8	0.433
Tang et al.	*ICASSP'24*	-	-	-	-	-	-	0.490	7.96	-	-	-	-
DocMamba	*ICMM'25*	-	-	-	-	-	-	0.529	7.07	0.338	-	-	-
Ours (direct)		0.591	9.51	0.247	0.769	494.1	0.473	0.531	7.25	0.335	0.153	423.4	0.463
Ours (average)		**0.593**	**9.47**	0.246	**0.717**	493.3	**0.475**	0.530	7.11	0.334	0.159	420.5	0.468

Figure 6. Qualitative comparison with deep learning-based methods on distorted images from several benchmarks. Columns 2-9 represent the dewarping results of each method.

better in aligning text lines and edges on some images but exhibits poorer quantitative metrics. Despite this, we have achieved competitive results on DocUNet. Overall, the comprehensive performance across three datasets demonstrates that TADoc delivers superior and robust results.

In addition, Tab. 1 also displays the model parameters and average inference speed of each method on DIR300. Due to our simple model architecture, streamlined pipeline, and method design for predicting sparse grid point warping flow, TADoc is able to achieve excellent results with minimal parameters. Even though the average method requires 20 steps of inference, parallel inference addresses this issue effectively.

Qualitative evaluation. We further compare the visual results of different dewarping methods side by side in Fig. 6. The document types of the selected images include flyers, posters, and magazines, while the deformation types involve rotation, bending, and folding. As depicted, previous methods lack robustness on various document types as well as distortion types and struggle with subtle deformation at the edge or corners, while our method works well on all these images. Our results not only excel in image edge restoration, but also achieve high-quality preservation of text paragraph structures and horizontal alignment of text lines. These improvements are particularly significant as our method does not rely on any prior knowledge of text lines or document layouts, unlike previous methods that

Table 3. Ablation study of timestep T and loss function on DIR300.

T	β	MS-SSIM ↑	LD ↓	AD ↓	DLS ↑
1	1	0.673	4.95	0.193	0.590
5	1	0.676	4.81	0.188	0.599
10	1	0.682	4.64	0.186	0.603
20	1	0.692	4.33	0.170	0.619
20	0.5	0.690	4.35	0.175	0.622
20	0.1	0.684	**4.29**	0.182	0.615
20	0	0.691	4.41	0.172	0.613
30	1	**0.694**	4.30	**0.166**	**0.627**

Figure 7. visualization of TADoc's restored result at various times as well as the final images obtained through the average method on real-world documents.

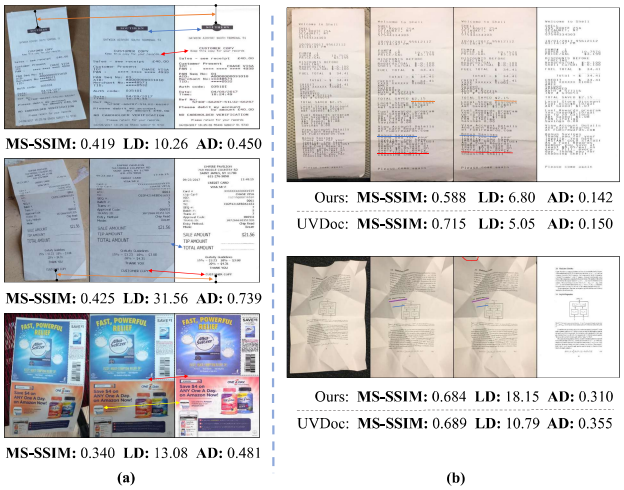

Figure 8. Bad case analysis on DocUNet. (a) From left to right: degraded image, TADoc dewarping image, and scanned image. (b) From left to right: degraded image, TADoc dewarping image, UVDoc dewarping image, and scanned image.

incorporated such priors.

As shown in the Fig. 7, we demonstrate TADoc's capability to restore various degrees of distortion at different times, illustrating the intermediate steps of our dewarping process. The last column presents the results of real-world images using the average method. It is evident that the average method provides more robust dewarping, particularly excelling in the restoration of text lines and edges. This highlights the superiority of our modeling, which allows for a more precise perception of the distortion extent, thereby achieving more robust dewarping results. Further visualization of DLS and more qualitative results on the benchmarks, and more analysis can be found in the appendix.

Ablation Studies. In ablation, we delve into the effects of total time steps T and loss function design, as well as the benefits our modeling method on DocTr. Specifically, different numbers of total time steps $T = 1, 5, 10, 20, 30$ are selected for model training respectively, and results are evaluated using the average method with corresponding T on the DIR300 benchmark. It is worth noting that the case of $T = 1$ degenerates TADoc to previous one-step modeling methods, which directly predicts the final warping flow of flat images from distorted images. As shown in Tab. 3, increasing the total number of time steps T can improve the effectiveness of model dewarping, yet the improvement becomes marginal beyond 20. Since the performance at $T = 20$ is already satisfactory, and considering the trade-off between effectiveness and training resources, we report the results with $T = 20$ for comparison. Next, we conduct an ablation study on the weight β of the reconstruction loss $\mathcal{L}_r$. Training TADoc with the L_1 losses on the backward map simply leads to a worse performance due to the lack of detailed guidance from images. Next, we conduct training on DocTr using our modeling method, with other settings kept consistent with the baseline, in order to validate the effectiveness of our method. As shown in the Tab. 4, our approach achieves significant improvement over the baseline on the DIR300 dataset, demonstrating the superiority of our modeling strategy.

Table 4. Ablation study of our modeling for DocTr on DIR300.

Method	MS-SSIM ↑	LD ↓	AD ↓
DocTr *(baseline)*	6.160	7.214	0.254
DocTr *(Ours)*	6.657 (↑ 8%)	5.509 (↓ 29%)	0.203 (↓ 20%)

4.5 Limitations

Despite the superior performance, limitations exist in our method. Firstly, we focus on the geometric rectification of the document, while the illumination rectification still needs to be addressed. Secondly, as a preliminary exploration, the dewarping process is formulated with linear interpolation out of intuition. We will explore a variation of interpolation in modeling and corresponding performance in the future.

5 Conclusion

In this work, we have rethought and modeled the document dewarping task to make it more in line with the laws of real physical scenes. To our knowledge, we are the first to introduce the concept of time steps in document dewarping tasks. Based on this, we propose a lightweight network termed TADoc with time embedding to predict intermediate processes of dewarping. Our method achieves state-of-the-art results on multiple benchmarks, with strong robustness confronted with various document types and deformation degrees. Besides, we introduce a novel metric DLS for comprehensively measuring geometric dewarping performance.

Acknowledgements

This work is supported by the National Natural Science Foundation of China (Grant NO 62376266 and 62406318).

References

[1] S. Biswas, A. Banerjee, J. Lladós, and U. Pal. Docsegtr: An instance-level end-to-end document image segmentation transformer. *arXiv preprint arXiv:2201.11438*, 2022.

[2] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam. Encoder-decoder with atrous separable convolution for semantic image segmentation. In *Proceedings of the European conference on computer vision (ECCV)*, pages 801–818, 2018.

[3] S. Das, K. Ma, Z. Shu, D. Samaras, and R. Shilkrot. Dewarpnet: Single-image document unwarping with stacked 3d and 2d regression networks. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 131–140, 2019.

[4] M. Diem, F. Kleber, and R. Sablatnig. Text classification and document layout analysis of paper fragments. In *2011 International Conference on Document Analysis and Recognition*, pages 854–858. IEEE, 2011.

[5] H. Feng, Y. Wang, W. Zhou, J. Deng, and H. Li. Doctr: Document image transformer for geometric unwarping and illumination correction. In *Proceedings of the 29th ACM International Conference on Multimedia*, pages 273–281, 2021.

[6] H. Feng, W. Zhou, J. Deng, Y. Wang, and H. Li. Geometric representation learning for document image rectification. In *European Conference on Computer Vision*, pages 475–492. Springer, 2022.

[7] A. Garz, M. Diem, and R. Sablatnig. Detecting text areas and decorative elements in ancient manuscripts. In *2010 12th international conference on frontiers in handwriting recognition*, pages 176–181. IEEE, 2010.

[8] R. Girshick. Fast r-cnn. In *Proceedings of the IEEE international conference on computer vision*, pages 1440–1448, 2015.

[9] M. Han and H. Li. Docmamba: Robust document image dewarping via selective state space sequence modeling. In *International Conference on Multimedia Modeling*, pages 304–318. Springer, 2025.

[10] K. He, X. Zhang, S. Ren, and J. Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 770–778, 2016.

[11] K. He, G. Gkioxari, P. Dollár, and R. Girshick. Mask r-cnn. In *Proceedings of the IEEE international conference on computer vision*, pages 2961–2969, 2017.

[12] X. Jiang, R. Long, N. Xue, Z. Yang, C. Yao, and G.-S. Xia. Revisiting document image dewarping by grid regularization. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 4543–4552, 2022.

[13] N. Journet, V. Eglin, J.-Y. Ramel, and R. Mullot. Text/graphic labelling of ancient printed documents. In *Eighth International Conference on Document Analysis and Recognition (ICDAR'05)*, pages 1010–1014. IEEE, 2005.

[14] P. Kumari and S. Das. Am i readable? transfer learning based document image rectification. *International Journal on Document Analysis and Recognition (IJDAR)*, pages 1–14, 2024.

[15] O. Lavialle, X. Molines, F. Angella, and P. Baylou. Active contours network to straighten distorted text lines. In *Proceedings 2001 International Conference on Image Processing (Cat. No. 01CH37205)*, volume 3, pages 748–751. IEEE, 2001.

[16] F. Lebourgeois, Z. Bublinski, and H. Emptoz. A fast and efficient method for extracting text paragraphs and graphics from unconstrained documents. In *11th IAPR International Conference on Pattern Recognition. Vol. II. Conference B: Pattern Recognition Methodology and Systems*, volume 1, pages 272–273. IEEE Computer Society, 1992.

[17] G. Li, H. Shen, and Y. Zhou. Beyond cropped regions: New benchmark and corresponding baseline for chinese scene text retrieval in diverse layouts. In *ICML*, 2025.

[18] J. Li, Y. Xu, T. Lv, L. Cui, C. Zhang, and F. Wei. Dit: Self-supervised pre-training for document image transformer. In *Proceedings of the 30th ACM International Conference on Multimedia*, pages 3530–3539, 2022.

[19] P. Li, W. Quan, J. Guo, and D.-M. Yan. Layout-aware single-image document flattening. *ACM Transactions on Graphics*, 43(1):1–17, 2023.

[20] Z. Li, Y. Shu, W. Zeng, D. Yang, and Y. Zhou. First creating backgrounds then rendering texts: A new paradigm for visual text blending. In *ECAI 2024*, pages 346–353. IOS Press, 2024.

[21] J. Lyu, W. Wang, D. Yang, J. Zhong, and Y. Zhou. Arbitrary reading order scene text spotter with local semantics guidance. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 5919–5927, 2025.

[22] K. Ma, Z. Shu, X. Bai, J. Wang, and D. Samaras. Docunet: Document image unwarping via a stacked u-net. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 4700–4709, 2018.

[23] A. Markovitz, I. Lavi, O. Perel, S. Mazor, and R. Litman. Can you read me now? content aware rectification using angle supervision. In *Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XII 16*, pages 208–223. Springer, 2020.

[24] G. Meng, Y. Wang, S. Qu, S. Xiang, and C. Pan. Active flattening of curved document images via two structured beams. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 3890–3897, 2014.

[25] L. Mischke and W. Luther. Document image de-warping based on detection of distorted text lines. In *Image Analysis and Processing–ICIAP 2005: 13th International Conference, Cagliari, Italy, September 6-8, 2005. Proceedings 13*, pages 1068–1075. Springer, 2005.

[26] Z. Qiao, Y. Zhou, D. Yang, Y. Zhou, and W. Wang. SEED: Semantics enhanced encoder-decoder framework for scene text recognition. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 13528–13537, 2020.

[27] H. Shen, X. Gao, J. Wei, L. Qiao, Y. Zhou, Q. Li, and Z. Cheng. Divide rows and conquer cells: towards structure recognition for large tables. In *Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence*, IJCAI '23, 2023.

[28] Y. Shu, W. Zeng, F. Zhao, Z. Chen, Z. Li, X. Yang, Y. Zhou, P. Rota, X. Bai, L. Jin, et al. Visual text processing: A comprehensive review and unified evaluation. *arXiv preprint arXiv:2504.21682*, 2025.

[29] H. Tang, J. Guo, T. Wang, Y. Yu, and C. Wang. Efficient joint rectification of photometric and geometric distortions in document images. In *ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 3690–3694. IEEE, 2024.

[30] Y.-C. Tsoi and M. S. Brown. Multi-view document rectification using boundary. In *2007 IEEE Conference on Computer Vision and Pattern Recognition*, pages 1–8. IEEE, 2007.

[31] F. Verhoeven, T. Magne, and O. Sorkine-Hornung. Uvdoc: Neural grid-based document unwarping. In *SIGGRAPH Asia 2023 Conference Papers*, pages 1–11, 2023.

[32] W. Wang, Y. Zhou, J. Lv, D. Wu, G. Zhao, N. Jiang, and W. Wang. TPSNet: Reverse thinking of thin plate splines for arbitrary shape scene text representation. In *Proceedings of the 30th ACM International Conference on Multimedia*, pages 5014–5025, 2022.

[33] G.-W. Xie, F. Yin, X.-Y. Zhang, and C.-L. Liu. Document dewarping with control points. In *Document Analysis and Recognition–ICDAR 2021: 16th International Conference, Lausanne, Switzerland, September 5–10, 2021, Proceedings, Part I 16*, pages 466–480. Springer, 2021.

[34] Z. Xu, F. Yin, P. Yang, and C.-L. Liu. Document image rectification in complex scene using stacked siamese networks. In *2022 26th International Conference on Pattern Recognition (ICPR)*, pages 1550–1556. IEEE, 2022.

[35] X. Yang, Z. Qiao, and Y. Zhou. Ipad: Iterative, parallel, and diffusion-based network for scene text recognition. *International Journal of Computer Vision*, pages 1–21, 2025.

[36] S. You, Y. Matsushita, S. Sinha, Y. Bou, and K. Ikeuchi. Multiview rectification of folded documents. *IEEE transactions on pattern analysis and machine intelligence*, 40(2):505–511, 2017.

[37] F. Yu and V. Koltun. Multi-scale context aggregation by dilated convolutions. *arXiv preprint arXiv:1511.07122*, 2015.

[38] F. Yu, Y. Xie, L. Wu, Y. Wen, G. Wang, S. Ren, X. Chen, J. Mao, and W. Li. Docreal: Robust document dewarping of real-life images via attention-enhanced control point prediction. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 665–674, 2024.

[39] W. Zeng, Y. Shu, Z. Li, D. Yang, and Y. Zhou. Textctrl: Diffusion-based scene text editing with prior guidance control. *Advances in Neural Information Processing Systems*, 37:138569–138594, 2024.

[40] J. Zhang, C. Luo, L. Jin, F. Guo, and K. Ding. Marior: Margin removal and iterative content rectification for document dewarping in the wild. In *Proceedings of the 30th ACM International Conference on Multimedia*, pages 2805–2815, 2022.

[41] L. Zhang, A. M. Yip, M. S. Brown, and C. L. Tan. A unified framework for document restoration using inpainting and shape-from-shading. *Pattern Recognition*, 42(11):2961–2978, 2009.

[42] W. Zhang, Q. Wang, K. Huang, X. Huang, F. Guo, and X. Gu. Document registration: Towards automated labeling of pixel-level alignment between warped-flat documents. In *Proceedings of the 32nd ACM International Conference on Multimedia*, pages 9933–9942, 2024.

[43] T. Zheng, Z. Chen, S. Fang, H. Xie, and Y.-G. Jiang. CDistNet: Perceiving multi-domain character distance for robust text recognition. *International Journal of Computer Vision*, 132(2):300–318, 2024.